

Introduction to Algorithms

Algorithms are step-by-step procedures for solving problems. They are fundamental to computer science, enabling tasks such as searching, sorting, and optimization. Understanding algorithms helps programmers write efficient and scalable code.

Key Points

1. Sorting Algorithms: Arrange data in order (e.g., QuickSort, MergeSort).
2. Searching Algorithms: Find elements in data (e.g., Binary Search).
3. Graph Algorithms: Solve network problems (e.g., Dijkstra's algorithm).

Example Code

```
numbers = [5, 2, 9, 1]
numbers.sort()
print('Sorted:', numbers)

def binary_search(arr, target):
    left, right = 0, len(arr) - 1
    while left <= right:
        mid = (left + right) // 2
        if arr[mid] == target:
            return mid
        elif arr[mid] < target:
            left = mid + 1
        else:
            right = mid - 1
    return -1
print('Index:', binary_search(numbers, 5))
```

Conclusion

Algorithms form the foundation of programming logic and problem solving.